

YASH GARG

DATA SCIENTIST | DATA ANALYST | AI/ML ENGINEER

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Summary

I am an experienced Data Scientist and Generative AI Engineer with **4.5+** years of expertise in designing, fine-tuning, and deploying advanced machine learning and open-source LLM-based solutions across financial, marketing, and customer analytics domains. Skilled in building scalable ML and Generative AI pipelines, including Retrieval-Augmented Generation (**RAG**) workflows, **LoRA** fine-tuning, and real-time model deployment using **FastAPI** and **Streamlit**. Proficient in Python, SQL, TensorFlow, PyTorch, and PySpark, with hands-on experience in cloud platforms such as **AWS**, **GCP**, and **Azure**. Strong background in **time-series** forecasting, recommendation systems, **NLP**, and data automation, delivering actionable insights and performance improvements in large-scale enterprise environments.

Skills

Programming Language	: Python, R
Query Language	: MySQL, Oracle, Hive, T-SQL, PL/SQL, MongoDB
Data Visualization	: Tableau, PowerBI, looker studio, Plotly, SeaBorn, Matplotlib, ggplot2
Data Wrangling and ETL	: Pandas, NumPy, PySpark, Apache Airflow, Azure Data Factory, Alteryx
Machine Learning	: Linear/Logistic Regression, Decision Trees, Random Forest, XGBoost, SVM, kNN, Gradient Boosting Machines, AutoML, Scikit-learn, TensorFlow, PyTorch, Keras, SparkML
Deep Learning	: CNNs, RNNs, LSTMs, GANs, Transformers (BERT, GPT)
Algorithm	: NLP, Time Series (ARIMA, SARIMA, Prophet)
Cloud Platform	: Azure (Azure ML, Data Factory, Synapse), AWS (SageMaker, S3, Redshift), GCP (BigQuery, Vertex AI)
Big Data Tools	: Hadoop, Spark, Kafka
Data Science Techniques	: Statistics, Hypothesis Testing, Causal Inference, Clustering, Feature Engineering
Version Control	: GIT
Project Management	: JIRA, Confluence

Work Experience

Data Scientist Jan 2025 – Till date

Y STEM and Chess Inc. | Boise, Idaho

- Designed machine learning models to forecast student performance trends, resulting in a **22%** improvement in learning outcomes through personalized Chess and STEM lesson plans.
- Automated the **delivery of actionable insights** to educators by building custom algorithms, leading to a **35% boost** in **instructional efficiency** and faster data-driven decisions.
- Built dynamic data pipelines and dashboards using **Python** and **Power BI** to track and visualize student progress, reducing manual analysis time by **40%**.
- Collaborated with educators to create a **machine learning-based recommender system** that guided individualized learning paths, enhancing early intervention accuracy for at-risk students.

Data Scientist (Research Analyst) Jan 2024 – Dec 2024

Georgia State University | Atlanta, Georgia

- Developed a Generative AI-powered **financial chatbot** using **Mistral-7B**, **LLaMA 2**, and **Falcon** models, fine-tuned with **LoRA** (Low-Rank Adaptation) for efficient domain-specific adaptation on financial datasets.
- Built a Retrieval-Augmented Generation (**RAG**) pipeline using **LangChain** and **FAISS** as vector database to ensure context-aware and factually consistent responses.
- Designed the chatbot user interface with **Streamlit**, enabling intuitive real-time interaction for financial queries.
- Deployed the chatbot within a secure **FastAPI** framework and optimized inference latency using quantization (4-bit) and GPU acceleration for real-time user interaction.
- Implemented prompt engineering and chain-of-thought reasoning strategies to enhance LLM interpretability and minimize hallucinations by 24% in test evaluations.
- Built and validated a time-series forecasting model using **ARIMA** with **96% accuracy** to predict material demand, enhancing procurement planning and reducing forecasting error in academic use cases.
- Conducted trend analysis and built custom dashboards using Power BI and Matplotlib after enriching datasets through automated web scraping with **BeautifulSoup** and **Selenium**.
- Queried and analyzed 100K+ records across multiple sources using **SQL** and **Spark** to extract behavioral insights from academic datasets, aiding research on financial literacy and product usage trends.

Data Scientist (Research Analyst) Aug 2023 – Dec 2023

Truist Bank | Atlanta, Georgia

- Built a Python-based benchmarking pipeline to compare Truist's TCFD (climate disclosure) metrics with national peers, leveraging NLP models (BERT, NER) and improving extraction accuracy by 30%.
- Applied **OCR technology** with **94% extraction accuracy** to automate document parsing and reduce **manual analysis time by 50%**.
- Collaborated with **ESG and sustainability teams** to define key disclosure metrics, ensuring alignment with **regulatory compliance** and industry reporting standards.

- Integrated the **NLP and OCR pipeline** into a **scalable reporting framework**, enabling **automated weekly summaries** for internal risk and compliance teams.
- **Developed interactive dashboards** using **Power BI** to visualize benchmarking results, enabling senior leadership to track ESG performance and gaps against competitors in real time.

Data Scientist

Jan 2022 – Jul 2023

- Infosys | Bangalore, India
- Led a 4-member team to enhance 10+ Customer Selection ML Algorithms and set up A/B tests for each enhancement using **Snowflake** and **PySpark** for scalable data processing on large datasets, boosting ad spend returns by 1.5x. Conducted Agile scrums for efficient project management using Trello.
 - Built **Conditional Spend ML Model** on skewed datasets for customer ranking algorithm, yielding **~2x** website traffic and **~1.5x** sales.
 - Applied **RNN, LSTM** on embedded products for Hyper-Personalized Recommendation System, achieving **40%** match-rate in recommendations.
 - Spearheaded **5+** A/B testing pipelines to measure marketing campaign KPI, with statistical analysis and tests to ensure significance.
 - Applied **causal inference** methodologies to analyze customer segments and campaigns to uncover actionable insights.
 - Collaborated with Marketing and Creative team to create 3 Customers Personas by utilizing customer data and applying Clustering.
 - **Automated** multiple ML Model and Customer Selection pipelines to **Apache Airflow**, reduced refresh time by **15 hours/week** and integrated ad-hoc Excel reports into **Tableau** dashboards, reducing manual effort by **20 hours/week**.
 - Employed BigQuery, **BQML**, **Kubernetes** (GCP) and **AWS S3** to clean, structure, and analyze distributed data for decision-making.

Data Analyst

Jan 2021 – Dec 2021

- Infosys | Bangalore, India
- Implemented and maintained interactive dashboards in **DOMO** and **Looker**, automating KPI tracking and increasing operational efficiency by **15%**.
 - Constructed **ETL** pipelines on **Alteryx** for SAP ERP to SAP HANA migration during a project's inaugural Go Live phase, documenting workflows in **Confluence**.
 - Developed and optimized **Power BI** dashboards and **DAX** queries, automating workflows that delivered **95%** accurate insights and improved reporting by **90%**.
 - Improved data retrieval across projects by optimizing **SQL** queries and database structures, cutting query execution time by **30%**.
 - Created **advanced SQL joins**, aggregations, subqueries, and Tableau dashboards, leading to **80%** improvement in business decision-making and report accuracy.

Education

- **Master's** in Data Science and Analytics from **Georgia State University**, Atlanta, Georgia

Aug 2023 – Dec 2024
- **Bachelor's** in Information Technology from **Guru Gobind Singh Indraprastha University**, Delhi, India

Aug 2016 – Nov 2020

Projects

- **Generative AI for synthetic finance data:** To tackle data privacy concerns while predictive algorithms, employed CT-GANs to create synthetic data which was validated against real data by pair-wise association and ML detection. Model created using synthetic vs real data scored AUC 0.81 vs 0.83.
- **Credit Cards Leads:** To capture potential credit card customers, researched multiple credit card customer case studies and utilized web scraping to create dataset that supplements the provided dataset by a major Consumer Bank. Through setting up a thorough Feature Engineering pipeline and by rigorous experimentation with algorithms, set-up a machine learning pipeline that evaluated f1 score >0.85 for multiple test sets.
- **Uber NYC Ride Data Analysis:** Conducted exploratory data analysis on real-world Uber ride data from New York City to identify temporal and geographic trends in rider demand. Utilized Python, along with libraries such as Pandas, NumPy, Matplotlib, and Seaborn, to perform data cleaning, feature transformation, and visualization. Extracted actionable insights into peak hours, day-wise usage patterns, and high-demand zones, supporting ride optimization strategies. Deployed findings in a Jupyter Notebook environment with reproducible plots and statistical summaries.

Certifications & Achievements

- Oracle Cloud: Oracle Cloud Infrastructure 2024 Generative AI
- AWS Certified Data Engineer – Associate: [Data Engineering](#)
- Best Data Analytics Poster at Georgia State University
- Infosys Star Performer for delivering effective solutions
- Managed a team for 4 associates when I had only 1 year of experience
- Deployed more than 5 models into production in a single financial year